Austin Konkel

Curriculum Vitæ austin.konkel@stonybrook.edu

Department of Mathematics, Stony Brook University, Stony Brook, NY 11794-3651

Education

Doctor of Philosophy in Mathematics May 2027 (Expected)
Stony Brook University, advised by Dennis Sullivan
Thesis: *Sheaves of groups defined by codimension one immersions*

Bachelor of Science in Mathematics 2021
Central Michigan University, advised by Debraj Chakrabarti and Meera Mainkar

International Baccalaureate Diploma 2017
International Academy of Macomb

Research Interests

Geometric topology, with a focus on immersions and low-dimensional manifolds

Publications and Preprints

Which immersions into Euclidean space are boundary immersions?
A. Konkel. Preprint, arXiv:2610.25618, 2026.

Bergman kernels of elementary Reinhardt domains
D. Chakrabarti, A. Konkel, M. Mainkar, and E. Miller. *Pacific J. Math.*, 306(1): 67–93, 2020.

Awards

Chairman's Award for Excellence in Service to the Department, University or Community 2026
Awarded by Stony Brook University, Department of Mathematics

Goldwater Scholarship 2020
Awarded by the Barry Goldwater Scholarship and Excellence in Education Foundation

Outstanding Mathematics Assistance Center Tutor Award 2020
Awarded by Central Michigan University, Department of Mathematics

Presentations

Seminar and Conference Talks

Geometry and Topology Seminar October 2026
"Inside Euclidean 3-space, which immersed 2-spheres are the boundary of some compact immersed 3-manifold?"
University of Massachusetts Amherst

Geometry, Dynamics, and Computer-Assisted Proofs Conference June 2026
"Sandy Blank – Which circles immersed generically into the plane bound an immersed disk?"
Heidelberg University

New York State Regional Graduate Mathematics Conference March 2026
"Sandy Blank – Which circles immersed generically into the plane bound an immersed disk?"
Syracuse University

BUGCAT Conference November 2025
"Sandy Blank – Which circles immersed generically into the plane bound an immersed disk?"
Binghamton University

Young Mathematicians Conference August 2019
"Bergman kernels of elementary Reinhardt domains"
Ohio State University

Summer Undergraduate Michigan Mathematics Research Conference July 2019
"Bergman kernels of elementary Reinhardt domains"
University of Michigan-Dearborn

Graduate Student Seminars

Graduate Student Seminar October 2026
"Inside Euclidean 3-space, which immersed 2-spheres are the boundary of some compact immersed 3-manifold?"
Stony Brook University

Graduate Student Seminar April 2026
"Sandy Blank - Which circles immersed generically into the plane bound an immersed disk?"
Stony Brook University

Graduate Student Seminar December 2024
"Immerse yourself into the world of Immersions"
Stony Brook University

Graduate Student Seminar February 2024
"One-dimensional dynamics should be trivial, right?"
Stony Brook University

Graduate Student Seminar October 2022
"Round Handles and Morse–Smale Systems"
Stony Brook University

Graduate Student Seminar October 2021
"Chip-Firing on Graphs"
Stony Brook University

American Mathematical Society Graduate Student Seminar November 2020
"Bergman kernels of elementary Reinhardt domains"
Central Michigan University

Teaching

Instructor

High School Lecturer for Mathematical Thinking Summers 2023 – 2026
Stony Brook University, Science and Technology Entry Program (STEP)

Lecturer for Math Camp: Introduction to Cryptography July 2025
Stony Brook University Math Camp

Lecturer for MAT 118: Mathematical Thinking Summer 2022
Stony Brook University

Teaching Assistant, Stony Brook University

MAT 122: Overview of Calculus with Applications Fall 2025

MAT 310: Linear Algebra Fall 2024

MAT 132: Calculus II Fall 2023

MAT 203: Calculus III with Applications Spring 2023

MAT 312: Applied Algebra Fall 2022

MAT 125: Calculus A Fall 2021

Grader, Stony Brook University

MAT 528: Foundations of Topology (Fall 2025); MAT 530: Topology, Geometry I (Fall 2024); MAT 314: Abstract Algebra II and MAT 360: Geometric Structures (Spring 2024); MAT 200: Logic, Language and Proof (Spring and Fall 2022)

Undergraduate Mentoring

Directed Reading Program, Stony Brook University

Eddy Ramirez: Exotic Spheres	Spring 2026
Maximillian Freund: Differential Topology	Spring 2023
Anatolii Vakaryuk: Measure Theory	Spring 2023

Independent Study

Jiaxuan Chen: Differential Topology	Summer 2026 – present
Anubrotaa Bhuniya: Differential Topology	Spring – Summer 2024

Service

Lunch Organizer for Dennis Sullivan's Stony Brook Seminar	Summer 2025 – present
Organizer of the Stony Brook Mathematics π-miles 5k Fun Run	March 2025 and 2026
Organizer of the 5/5/25 Stony Brook Mathematics Square Root Day Fun Run	May 2025
Organizer for the Stony Brook Graduate Student Employees Union	2022 – 2024
President of Central Michigan Kappa Mu Epsilon Mathematics Society	2020 – 2021
Vice President of Central Michigan Kappa Mu Epsilon Mathematics Society	2019 – 2020

Employment

Teaching and Research Assistant, Stony Brook University	2021 – present
Mathematics Grader and Tutor, Central Michigan University	2018 – 2021
Automotive Parts Laborer, Complete Prototype Services	2018 – 2019
Dishwasher and Prepcook, Little Joe's Coney Island	2017

Other Interests

Camping and being outdoors

Painting and assembling miniatures

Playing board games and competing in tournaments

References

Dennis Sullivan — Advisor
Stony Brook University and CUNY Graduate Center · sullivan0212@gmail.com

Maggie Miller — Research
University of Texas at Austin · maggie.miller.math@gmail.com

William Goldman — Research
University of Maryland · wmg@umd.edu

Christine Veloso — Teaching
Science and Technology Entry Program, Stony Brook University · christine.veloso@stonybrook.edu